

6DATA005W.2 Operational Research and Optimisation Team Project (2026)

Module leader	Fang He	
Units	Team Project	Individual report
Weighting:	30%	30%
Qualifying mark	30%	30%
Learning Outcomes Covered in this Assignment:	LO1 Demonstrate a systematic understanding of the principles and techniques for business analytics and decision making; LO2 Demonstrate critical understanding and assessment of optimisation and analytical models to analyse a real-world problem in a business context, and to develop a comprehensive solution; LO3 Apply appropriate software and/or programming as an aid to produce solutions; LO4 Interpret and evaluate comprehensively model solutions, assess critically their usefulness for decision-making and make sound recommendations.	
Handed Out:	28 January 2026	
Due Date	9 March 2026, 1:00pm presentation slides submission to Blackboard 10 and 11 March 2025, Group presentation during tutorial time	23 April 2026, 1:00pm, Individual report submission to Blackboard
Expected deliverables	Oral presentation and slides (9 March 2026), Report as pdf or MS Word file (23 April 2026), model files (23 April 2026)	
Method of Submission:	Online via Blackboard	
Type of Feedback and Due Date:	Written feedback and marks 15 working days (3 weeks) after the submission. All marks will remain provisional until formally agreed by an Assessment Board.	

Copying and plagiarism

Any external sources utilized should be correctly referenced using a common referencing technique (e.g., the Harvard technique). For more details on referencing please visit <http://www.westminster.ac.uk/study/current-students/support-and-facilities/library-it-services/referencing>.

Copying and plagiarism carry severe penalties. Please note that the University offers an online learning tutorial designed to help students understand and avoid plagiarism. This can be accessed by any student under My Organisation on Blackboard. The tab is labelled 'Avoiding Plagiarism'.

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, to minimum of the pass mark (40%), as a penalty for late submission. If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Faculty Registry Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website: <http://www.westminster.ac.uk/study/current-students/resources/academic-regulations>

Coursework Description

Everyone makes decisions every day. These decisions can range from trivial choices, like what to eat for lunch, to significant ones, such as deciding which university to attend. At the organisational level, it is critical to determine how to allocate limited resources—both materials and human capital—to achieve optimal performance.

This module aims to equip you with methods for identifying and applying analytical skills to a range of problems. To bring these concepts to life, you will work on a group project to model and solve a real-world problem using your analytical expertise. You will choose **one of two provided case studies** and apply the concepts learned throughout the module to define the problem, build a model, and develop a solution.

Task 1: Problem Statement. Clearly articulate the problem selected for your team project. Briefly explain the nature of the problem, identify the beneficiaries in the case study, and outline the objectives and constraints in plain terms.

Task 2: Input Data Handling. Identify the input data required for the problem, i.e., the information you already know. Use notations to represent this data, which will serve as the input parameters for your model.

Task 3: Model Building. Using the insights from the previous tasks, formulate the problem as an Integer Programming (IP) model. Clearly define the decision variables, objective function, and constraints.

These tasks will be presented in a group presentation. You are encouraged to leverage discussions and idea exchanges within your group to achieve the best outcome. You need to be able to answer the questions from your marker/tutor to show your understanding. Feedback will be provided to improve your model.

Your improved model will be included in your individual report. In your individual report, you will continue by exploring the use of appropriate tools to implement the model developed and presenting your solutions (**Task 4**). Additionally, you are expected to interpret and comprehensively evaluate the model's solutions (**Task 5**). Your report should conclude with a reflection on the usefulness of your model and solution method for decision-making, as well as personal insights gained from the project (**Task 6**).

Your personal reflection could include one or two mistakes made during the project and the lessons learned from them. Ideally, these mistakes should have long-term implications for your growth, such as learning how to create an effective model, understanding the complexity of developing a new solution technique, or recognising the time required to complete certain tasks.

Deliverable 1: Group Project Presentation

- **Date:** 10 and 11 March 2026 (during tutorial time)

- **Submission Deadline:** Slides must be submitted to Blackboard by 13:00 on 9 March 2026.
- **Details:** Each group will give a 5–10-minute presentation during Week 8. PowerPoint slides can be used, and the presentation should cover **Tasks 1–3** outlined previously. You will receive feedback from your peers and the tutor to improve your model for your individual written report.

Deliverable 2: Individual Report

- **Submission Deadline:** 23 April 2026 by 13:00
- **Structure:**
 - **Part 1:** written report based on **Tasks 1–3** from the group presentation. This section should begin with a brief problem description, followed by a literature review on the problem and its domain, and your improved model based on the feedback received from your tutor. Please note as this is an individual report you must write this part up individually, but the model you developed as a group could be shared.
 - **Part 2:** Covers **Tasks 4–6** described above.
- **Report Requirements:**
 - Original work (no plagiarism).
 - 1800-2000 words, including tables, figures, and references but excluding appendices.
 - Charts, small tables, and graphs can be included within the text, while large tables, spreadsheets, and any other detailed model analysis should be added as appendices.
- **Additional Submission:** A single ZIP file containing all data and models developed also needs to be submitted to Blackboard.

Limits to Excel Solver and Alternatives

Solver is limited to 200 decision variables for all models, and it is limited to 100 constraints for nonlinear models. While many successful projects will be possible with these limits, other projects will need software capable of solving much larger optimisation problems. There are also tools that permit upfront programming or scripting to formulate the model. You are encouraged to learn and use other tools (such as OpenSolver, CPLEX, AMPL®, MATLAB™, open source solvers in Python, etc).

Team Project Marking scheme

Your group presentation will be marked as follows (in total of 100):

Presentation (20 marks in total)

Clear presentation, regardless of content	5 marks
Good time keeping and pace throughout	5 marks
Appropriate use of visual aids on slides	5 marks
Well-structured slides	5 marks

Content: understanding and formulation (80 marks in total)

Clear understanding and description of the problem domain and problem	10 marks
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clearly described input data handling and definition with clear notations	20 marks
Correct and clearly explained problem formulation with decision variables, objective functions and constraints that define the problem	30 marks
Be able to answer the questions and show good understanding of the model	20 marks

Your individual coursework report will be marked based on the following indicative marking criteria (in total of 100):

Problem identification and description. Identify the problem you are interested in then describe it clearly. Research on background of the problem and similar problems; show a clear understanding of problem domain and problem.	10 marks
Literature review. A critical literature review on models and methods that are applied to solve your problem and/or related problems.	10 marks
Input data handling.	10 marks
Problem modelling. Sound and well-justified mathematical/algebraic problem formulation; clearly explained decision variables (3 marks), objective functions (4 marks) and constraints that define the problem (18 marks).	25 marks
Problem solving. Evidence of ability to apply solution approaches to the problem. Describe experiments that you ran with the model.	35 marks
Evaluation of the solution. Good analysis and interpretation of results.	5 marks
Reflection.	5 marks

Case file 1- Sophie's module selection

Sophie is a third-year student taking Finance course at the University of London. After the module fair, she was told she has five more modules she needs to take: Business Strategy (BS101), International Finance (FIN 300), one industry-based module, and any two finance optional modules. An industry-based module is a requirement at the university that has an industry internship component. Many of the industry-based modules are offered by the School of Computer Science and Sophie would like to take one of those. In particular, two modules she finds interesting are Business Computing (CS 101) and Web Design for Nonprofit Organizations (CS 102). After looking at the finance optional module choices, she noticed four potential finance modules: Data Analysis in Finance (FIN 315); Risk Management (FIN 316); Options, Futures and Swaps (FIN 317); and Fixed Instruments and Markets (FIN 318). Sophie would like to avoid morning classes because her parttime job requires her to work a few hours mostly weekday mornings.

When she selects the modules for her third year, she would like to keep in mind her priorities. Her priorities are first, the content of the module, second, the reputation of the instructor, and third, the timing of the module. She decided she will assign a rating between 1 and 5 to each module section under consideration. From the online class schedule, she has made a list of module sections offered, as shown in Table 1. All the modules have at least two sections to be chosen from. Some sections meet once a week for three hours, and some meet twice a week alternating between one-hour and two-hour periods. An "hour" at the University is 55 minutes long.

To rate the module sections, Sophie considered three factors: content, instructor, and timing. The rating is the weighted average of the three factor ratings. She rated the content of the module based on her interest in it, using a scale from 1 (poor) to 5 (extremely interested). The reputation of the instructor is also a value from 1 to 5, coming from published student comments and word of mouth from classmates. The timing of the module is also rated from 1 to 5 according to her own preference. The rating of the module is provided in the final column of Table 1. Sophie wonders how to use this information to obtain a good selection of modules for her third year.

The task of this case is to create a good selection of the modules for Sophie. Data file for Case 1.xlsx is provided on Blackboard Assessment Section.

Table 1 Available module Data

Module	Time	Section	Rating
BS101	Business Strategy	M 6–8:45 P.M.	4.3
BS101	Business Strategy	T 6–8:45 P.M.	3.8
BS101	Business Strategy	W 6–8:45 P.M.	3.5
BS101	Business Strategy	F 6–8:45 P.M.	3.5
BS101	Business Strategy	M 1:25–2:20 P.M.	4.6
BS101	Business Strategy	W 1:25–3:15 P.M.	2.7
		T 1:25–3:15 P.M.	
FIN 300	International Finance	Th 1:25–2:20 P.M.	3.3
		W 6–8:45 P.M.	
		T 1:25–3:15 P.M.	
CS101	Intergenerational Computing	W 2:30–5:15 P.M.	4.4
CS101	Intergenerational Computing	Th 2:30–5:15 P.M.	3.1
CS102	Web Design for Nonprofit Organizations	T 6–8:45 P.M.	3.7
CS102	Web Design for Nonprofit Organizations	W 2:30–5:15 P.M.	3.5
FIN 315	Data Analysis in Finance	Th 6–8:45 P.M.	3.0
FIN 315	Data Analysis in Finance	M 1:25–2:20 P.M.	3.7
		W 1:25–3:15 P.M.	
FIN 316	Risk Management	M 6–8:45 P.M.	3.6
FIN 316	Risk Management	M 1:25–3:15 P.M.	3.9
		W 1:25–2:20 P.M.	
FIN 317	Options, Futures and Swaps	T 6–8:45 P.M.	3.2
FIN 317	Options, Futures and Swaps	T 1:25–3:15 P.M.	3.4
		Th 1:25–2:20 P.M.	
FIN 318	Fixed Instruments and Markets	M 6–8:45 P.M.	3.0
FIN 318	Fixed Instruments and Markets	W 6–8:45 P.M.	3.5

Case 2- Canton Fair event scheduling

The Canton Fair or China Import and Export Fair, is a trade fair held in the spring and autumn seasons each year since the spring of 1957 in Canton, Guangdong, China. It is the oldest, largest, and the most representative trade fair in China. This case study simplifies and tried to mimic the scheduling of the Fair.

The Fair organizes trade events around the world to provide a forum for international sellers to meet international buyers of varieties of products. During these events, sellers are stationed in booths and are visited by buyers according to prescheduled appointments. Because of the limited time slots available in each event and the fact that the number of buyers and sellers can be quite large, Canton Fair attempts to schedule the seller-buyer appointments in advance of the event in a manner that maximizes preferences. The idea is to match mutual interests to produce the most effective use of available time slots(sessions) during the event. The input data of the problem is the buyers, the sellers, and the pre-scheduled time slots(sessions).

The basic restriction of the schedule includes: Each buyer or seller can meet at most once per session and each specific buyer-seller pair meeting can take places at most one session.

Buyer and seller's meeting preferences are determined as following: Following the registration of all buyers and sellers, each seller provides Canton Fair a prioritised list of buyers whom the seller wants to see. A similar list is demanded of each buyer with respect to sellers. The list assigns the value 1 to the top choice, with larger values implying lower preferences. Those lists need not be exhaustive, in the sense that sellers and buyers are free to express interest in meeting with some but not all registered counterparts. For example, in a list with 100 sellers, a buyer may seek meetings with 10 sellers Only, in which case the expressed preferences will be 1,2, ..., 10 for the selected sellers.

The raw data gathered from the buyers/sellers list may then be expressed algebraically as

b_{ij} = ranking assigned by buyer i to a meeting with seller j

s_{ji} = ranking assigned by seller j to a meeting with buyer i

B = maximum number of preferences elected by all buyers

S = maximum number of preferences elected by all sellers

α = relative weight of buyer preferences, $0 < \alpha < 1$

$1 - \alpha$ = relative weight of seller preferences

Based on these definitions, the buyer-seller preferences can be calculated as

$$c_{ij} = \begin{cases} 1 + \alpha \left(\frac{B - b_{ij}}{B} \right) + (1 - \alpha) \left(\frac{S - s_{ji}}{S} \right), & \text{if } b_{ij} \neq 0 \text{ and } s_{ji} \neq 0 \\ 1 + \alpha \left(\frac{B - b_{ij}}{B} \right), & \text{if } b_{ij} \neq 0 \text{ and } s_{ji} = 0 \\ 1 + (1 - \alpha) \left(\frac{S - s_{ji}}{S} \right), & \text{if } b_{ij} = 0 \text{ and } s_{ji} \neq 0 \\ 0, & \text{if } b_{ij} = 0 \text{ and } s_{ji} = 0 \end{cases}$$

The logic behind these formulas is that a smaller value of b_{ij} means a higher value of $B - b_{ij}$. Hence, a higher score assigned to a requested meeting between buyer i and seller j . A similar interpretation is given to the score $S - s_{ji}$ for a seller's requested meeting with buyer i . Both scores are normalised to values between 0 and 1 by dividing them by B and S , respectively, and then they are weighted by α and $1 - \alpha$ to reflect the relative importance of the buyer and seller preferences. Note that $b_{ij} = 0$ and $s_{ji} = 0$ indicate that no meetings are requested between buyer and seller.

The task of this case is to create a buyer-seller pair meeting schedule to maximise their preferences. Data file for Case 2.xlsx is provided on Blackboard Assessment Section.



The end